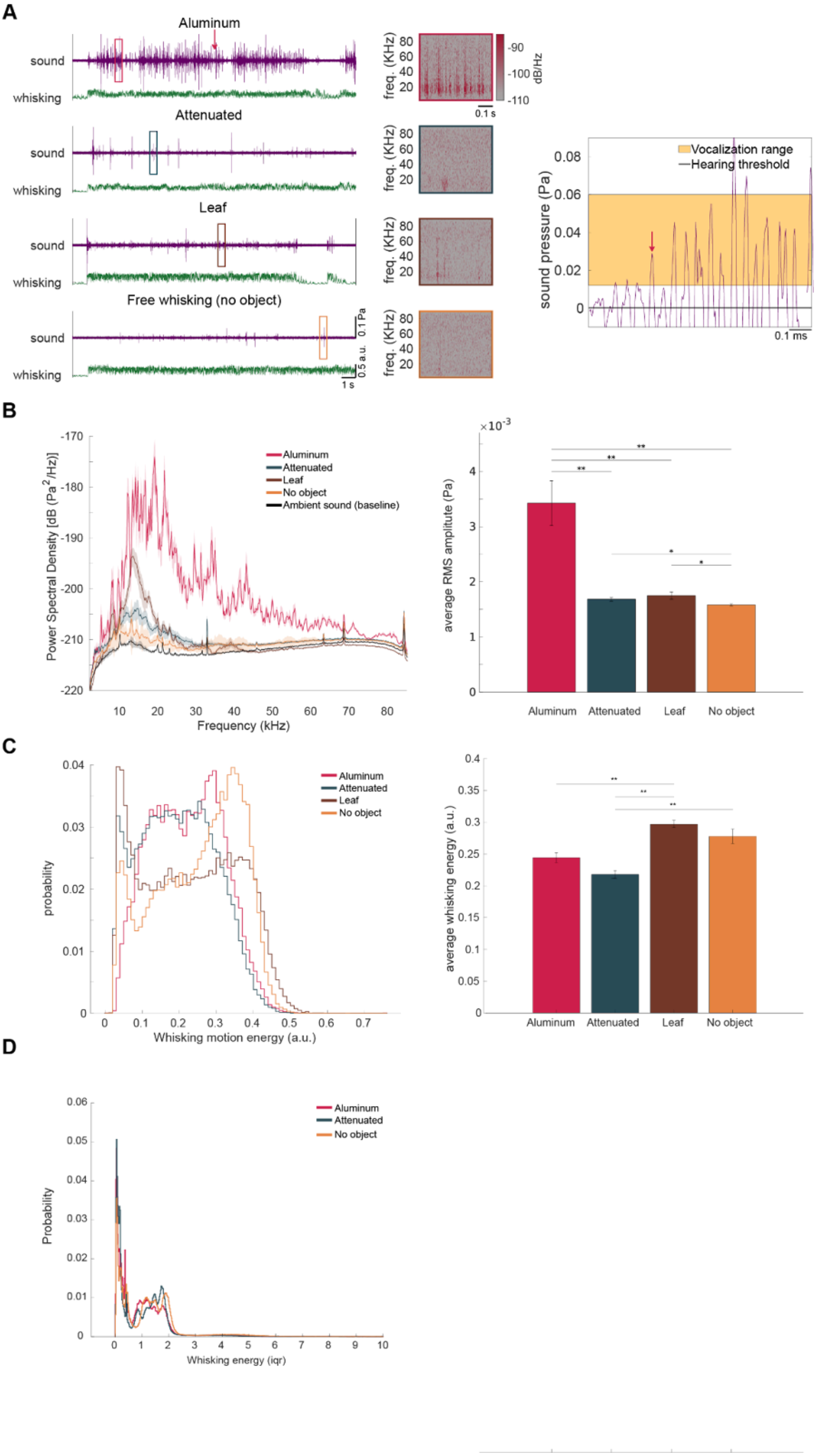
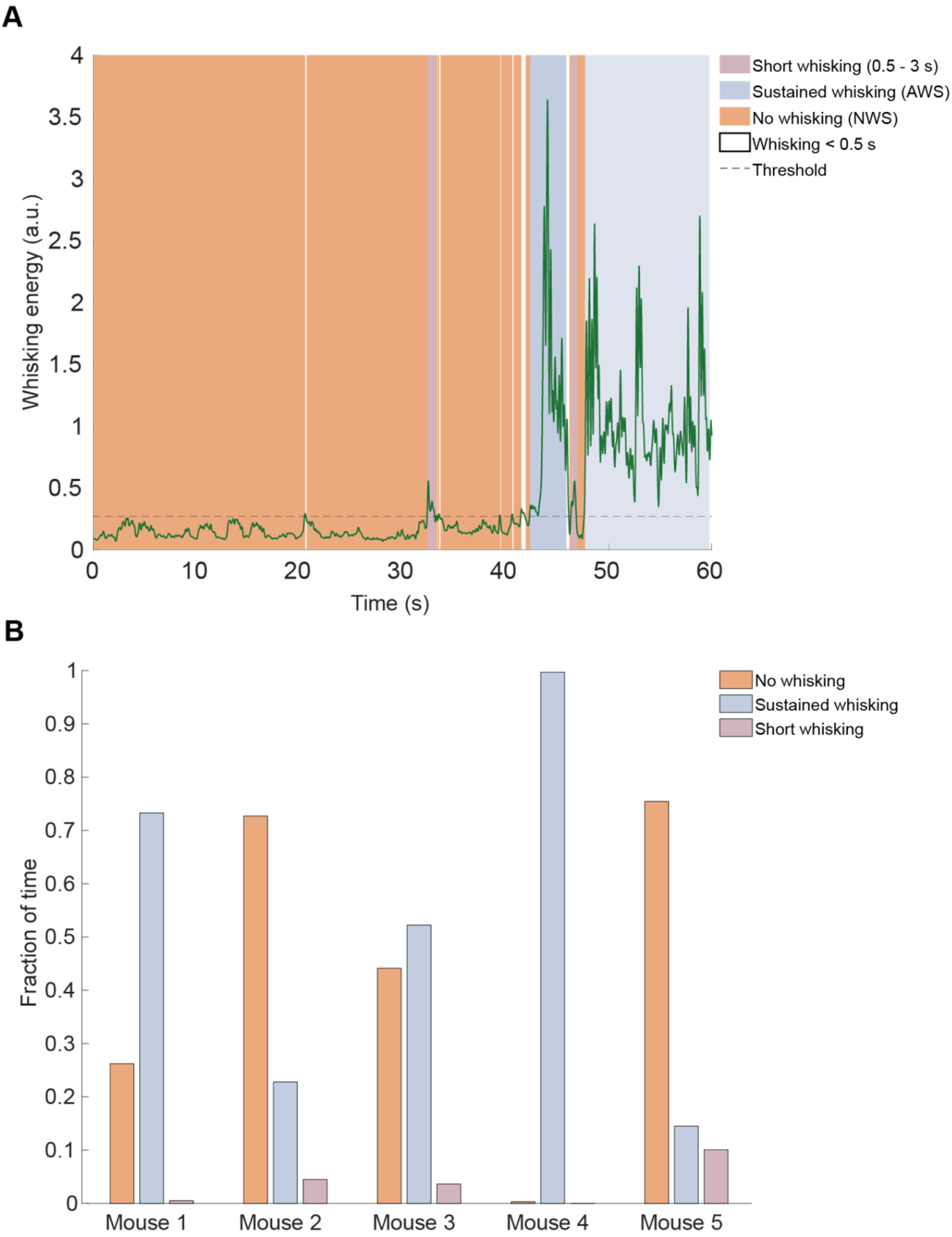


A
Aluminum
sound
whisking
freq. (KHz)
80
60
40
20
-90
-100
-110
dB/Hz
0.1 s
Attenuated
Leaf
Free whisking (no object)
0.1 Pa
0.5 a.u.
1 s
Vocalization range
Hearing threshold
sound pressure (Pa)
0.08
0.06
0.04
0.02
0
0.1 ms
B
Power Spectral Density [dB (Pa^2/Hz)]
-170
-180
-190
-200
-210
-220
Frequency (kHz)
Aluminum
Attenuated
Leaf
No object
Ambient sound (baseline)
×10^-3
average RMS amplitude (Pa)
Aluminum
Attenuated
Leaf
No object
C
probability
Whisking motion energy (a.u.)
average whisking energy (a.u.)
D
Probability
Whisking energy (iqr)
Aluminum
Attenuated
No object

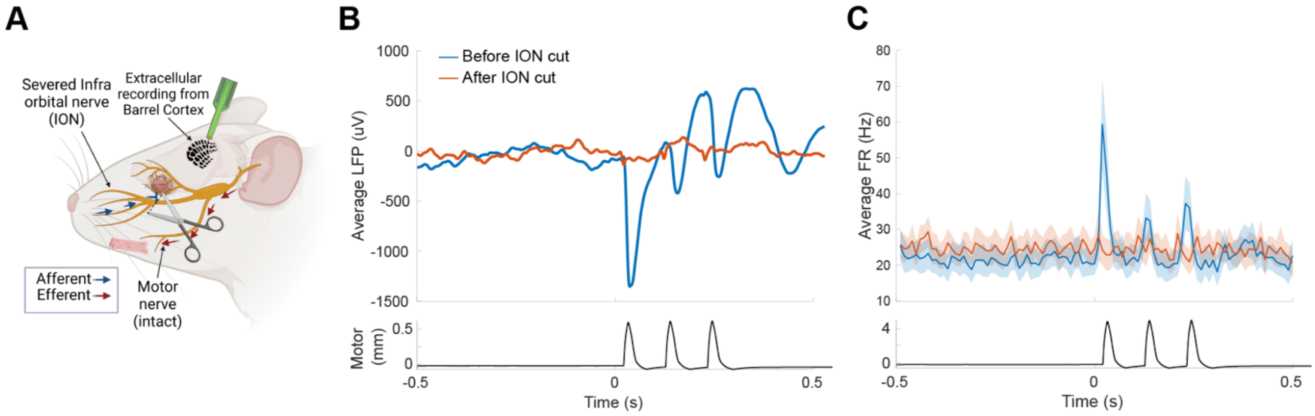
Supplementary Figure 1 – The pattern of whisking is similar for different objects.

(A) Comparison of whisking sounds with different objects: aluminum foil, attenuated foil, dry leaf, and no object. Colored boxes show the spectrogram of the matching-colored box in the trace. Right, zoom in on a section from the aluminum trace indicated by the red arrow. Comparison to the mice hearing threshold and the range of mouse vocalization (shaded yellow).
(B) Left, power spectrum density of the sounds during whisking epochs with different objects and comparison to the room ambient noise (black). Right, The average RMS amplitude of the sounds during whisking epochs for each object.
(C) Left, distribution of whisking motion energy (see Methods) for the different objects for the mouse in the example, including the dry leaf. Right, average whisking energy across the whisking epochs.
(D) A distribution of all the whisking motion energy from all the mice presented in Figure 1, normalized to the median from each mouse.
Mean ± 95% CI. *p < 0.05, **p < 5e-5 . In all panels: pink, aluminum; gray, attenuated; yellow, no object; brown, dry leaf.



Supplementary Figure 2 – Identifying whisking epochs and Quantification of whisking types

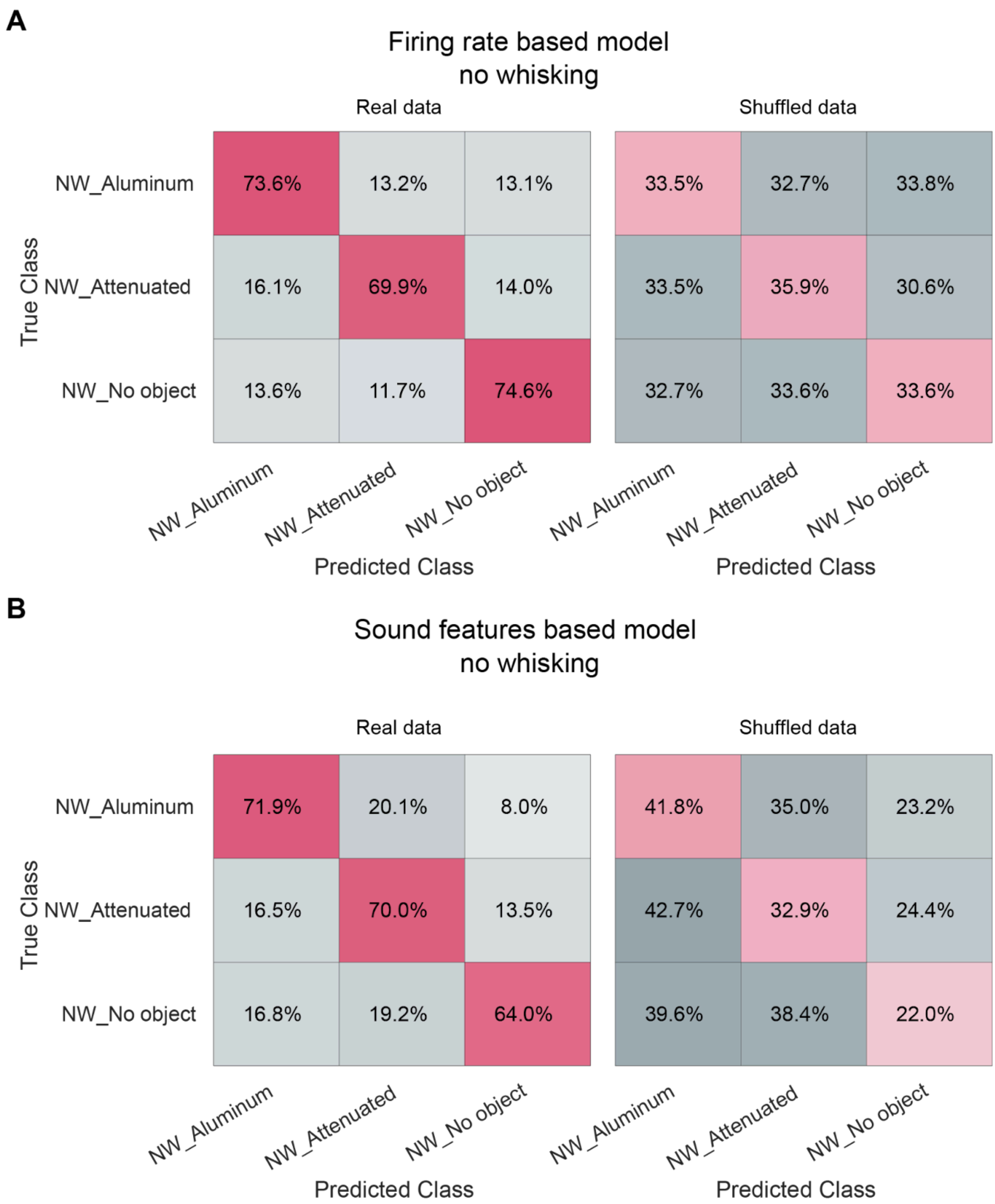
(A) An example of how whisking epochs were identified and categorized based on whisking motion energy.
(B) Quantification of the time spent in each whisking epoch (no whisking, short whisking, and sustained whisking in the analysis).



Supplementary Figure 3 – Cutting the ION successfully abolishes activity in the barrel cortex in response to whisker stimulation

(A) Recording from the barrel cortex of anesthetized mice during whisker deflection. The ION conveys the sensory information from the whiskers to the CNS, cutting it in the indicated site abolishes sensation while leaving the motor nerve and movement intact.
(B) LFP before and 20 minutes later, after ION cut in response to three consecutive whisker deflections. N = 32 electrodes.
(C) Same as (B) but for isolated units. Before ION cut, n = 29. After ION cut n = 29.

Mean ± 95% CI.



Supplementary Figure 4 – Classification of objects in the non-whisking state (NWS).

Classification based on neuronal firing and sound features was achieved with the random forest classifier, the same as in Figure 4 (B), but for models trained and tested only on data from the NWS periods and labeled based on the presented object.